Shivanshu Pandey

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EDUCATION

Indian Institute of Technology Hyderabad 2025 – 2027
Masters of Science, Physics *(Currently in 3rd Semester)*

Core Courses: High Enegry Particle Astrophysics, Plasma Physics, Particle Physics, Classical Mechanics, Quantum Mechanics I & II, Electrodynamics, Mathematical Physics I & II, Statistical Mechanics, Computational Physics, Optics & Photonics, Atomic & Molecular Physics.

VBS Purvanchal University 2020 – 2023
Bachelor of Science (Physics, Chemistry, Mathematics)

Core Courses: Calculus, Linear Algebra, Differential Equations, Real & Complex Analysis, Numerical Analysis & C Programming, Optics & Lasers, Electronics, Solid State & Nuclear Physics.

PROJECTS

Cosmic Ray Propagation in the Galaxy
- Developed 1D, 2D, and 3D simulations of cosmic ray transport from supernova sources through the interstellar medium
- Used Monte Carlo methods to model pitch-angle scattering and random walk of 1500+ particles
- Solved the diffusion PDE using the Crank–Nicolson method with the Thomas algorithm
- Validated results via statistical comparison with analytical solutions
- Applied Dirichlet boundary conditions to study galactic escape and particle residence times

Quantifying Sunspots using SAOImage DS9
- Analyzed solar FITS images using SAOImage DS9 to identify and study sunspots.
- Measured sunspot sizes and estimated their physical areas using image analysis techniques.
- Gained experience in astronomical data analysis and handling observational data.

Optical Spectrometer
- Designed and constructed a low-cost optical spectrometer using a diffraction grating and webcam-based detection
- Built optical system including slit mechanism, light isolation enclosure, and precise alignment setup
- Captured and analyzed spectral data using image processing techniques
- Demonstrated wavelength separation and intensity measurement of different light sources
- Estimated temperature of the Sun from observed solar spectrum using Wien's displacement law

8-bit Breadboard Computer
- Worked on the design and implementation of an 8-bit breadboard computer using digital logic components.
- Built and tested core modules including logic gates, registers, counters, and memory circuits.
- Gained hands-on experience with computer architecture, digital electronics, and instruction execution.

POSITIONS OF RESPONSIBILITY

Class Representative of MSc Physics, IIT Hyderabad July 2025 - Present
Core Member of Cepheid - Astronomy Club June 2026 - Present
Core Member of Nakshatra Astronomy Club Aug 2026 - Present
Volunteered for National Science Week, IIT Hyderabad
Volunteered for National Space Week, IIT Hyderabad

ACHIEVEMENTS

IIT JAM Physics 2025 Qualified AIR 514
Shortlisted for interview for **iPhD** programs at IISER Pune, IISER Bhopal, and NISER Bhubaneswar.

EXPERIENCE & ACTIVITIES

Geo-processing Using Python July 2023 — 12 Hours IIRS (ISRO)
- Completed a training program on Python-based geospatial data processing and remote sensing, gaining exposure to Earth observation data analysis and geoprocessing techniques.

SKILLS

Programming: Python
Libraries: NumPy, Matplotlib, SciPy, Pandas
Software: DS9, TOPCAT
Tools: GitHub, VS Code
Documentation: LaTeX